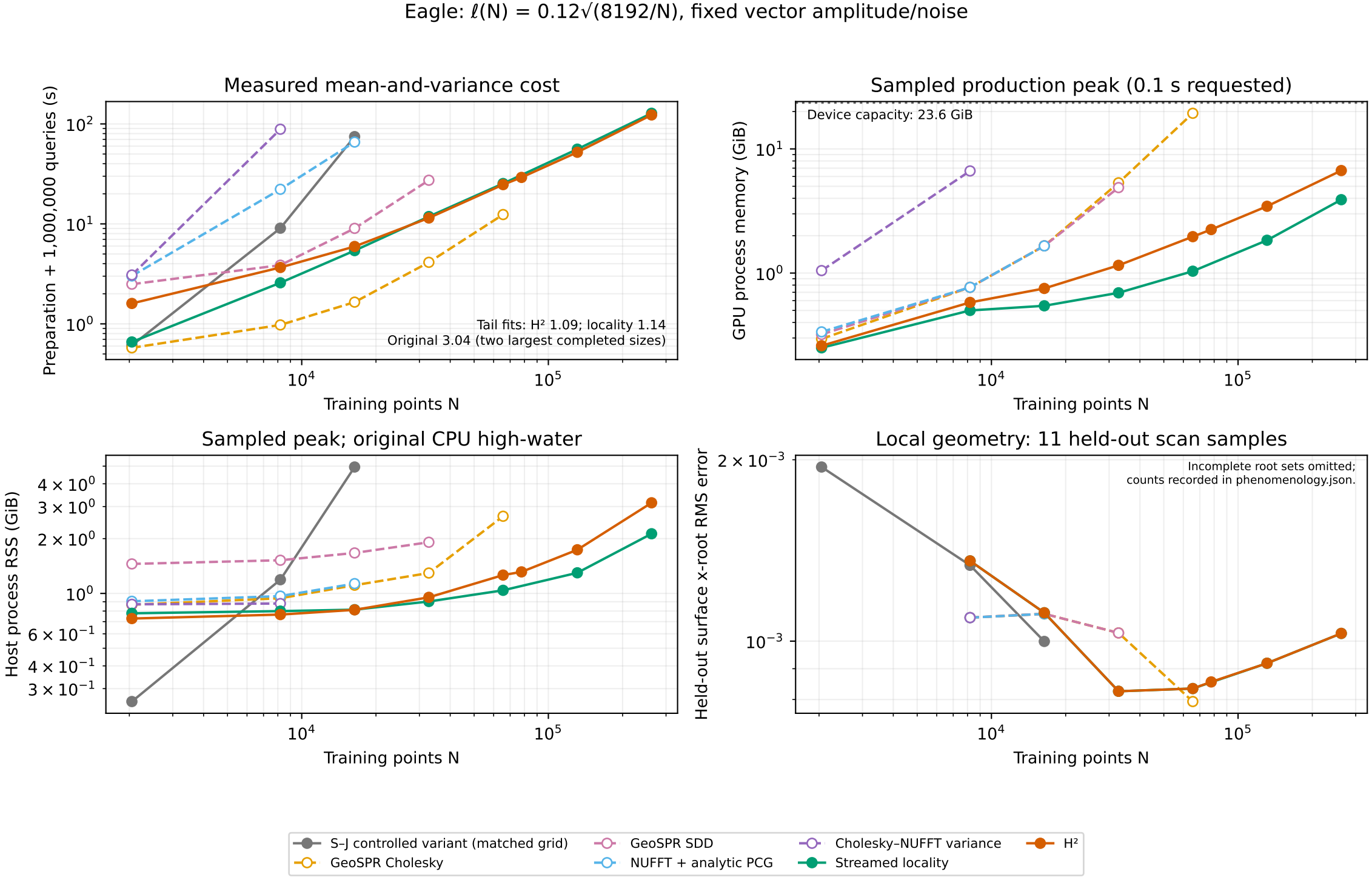
Eagle: ℓ(N) = 0.12√(8192/N), fixed vector amplitude/noise
Measured mean-and-variance cost
Preparation + 1,000,000 queries (s)
Tail fits: H² 1.09; locality 1.14
Original 3.04 (two largest completed sizes)
Training points N
Sampled production peak (0.1 s requested)
GPU process memory (GiB)
Device capacity: 23.6 GiB
Training points N
Sampled peak; original CPU high-water
Host process RSS (GiB)
Training points N
Local geometry: 11 held-out scan samples
Held-out surface x-root RMS error
Incomplete root sets omitted;
counts recorded in phenomenology.json.
Training points N
S–J controlled variant (matched grid)
GeoSPR SDD
Cholesky–NUFFT variance
H²
GeoSPR Cholesky
NUFFT + analytic PCG
Streamed locality
Hollow/dashed: periodic finite-band solver accuracy only. ×: rejected numerical settings. Cross-model calibration unresolved.